

Estimation Theory

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1 Introduction

Definition 1.1 (Population). A population is the collection of a set of all measurements that are of a particular interest.

Example. A simple example can be perhaps, the set of *height measurements* of some large group of people.

Remark. In the above example, the large group of people themselves do not constitute a population instead it is their height measurements is what we call a population.

Definition 1.2 (Sample). A sample is a subset of the population.

Example. Following our above example for population, a sample can be a set of height measurements of some *subset* of the corresponding large group of people.

In this part of the course we are interested in *estimating* some *parameters* associated with a population based on a sample that we have collected. This is what we call a **Statistical Inference**. Unless otherwise specified, we will be working with random sample, as defined below.

Definition 1.3 (Random Sample). Let N and n represent the number of elements in the population and sample respectively. A random sample of size n is a sample collected in such a way that each of the $\binom{N}{n}$ samples has an equal probability of being selected.

The following few results which we have discussed quite often in the class will be very useful throughout this course.

Theorem 1.1 (Linearity of Expectation). *For any two random variables X and Y , the linearity of expectation tells us that,*

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

In more generality, this is stated as follows. Let $X_1, X_2, \dots, X_n$ be a sequence of random variables. Then,

$$\mathbb{E} \left[\sum_{i=1}^n (X_i) \right] = \sum_{i=1}^n \mathbb{E}[X_i]$$

Theorem 1.2. *If $X_1, X_2, \dots, X_n$ are independent random variables, then we have,*

$$\mathbb{E} \left[\prod_{i=1}^n X_i \right] = \prod_{i=1}^n \mathbb{E}[X_i]$$

Theorem 1.3. *If $X_1, X_2, \dots, X_n$ are independent random variables, then we have,*

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i)$$

2 Fundamentals of Estimation

Definition 2.1 (Statistic). A statistic is a function of the observable random variables in a sample and some known constants.

Example. Suppose $X_1, X_2, \dots, X_n$ are iid random variables. The the following value, $\bar{X}$, called sample mean, which we will see in more detail shortly, defined as,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is an example of a statistic.

Observe that a statistic is also a random variable (Why?), thus it has its own probability distribution as well, which is called a **sampling distribution**.

Exercise 2.1. A fair die is rolled thrice. Let X_i be a random variable that denotes the outcome of the i_{th} roll. Find sampling distribution of $\bar{X}$, where,

$$\bar{X} = \frac{1}{3} \cdot (X_1 + X_2 + X_3)$$

We can have two types of estimate, one being, *Point Estimate*, defined as,

Definition 2.2 (Point Estimate). A point estimate gives a single point value as an estimate for some target parameter.

and the other being, *Interval Estimate*, defined as,

Definition 2.3 (Interval Estimate). An interval estimate gives an interval as an estimate for some target parameter.

We use the so-called *estimator* to estimate some parameter which is defined as follows.

Definition 2.4 (Estimator). An estimator is a rule or more generally some formula that tells us how to estimate some parameter.

Definition 2.5. Suppose θ denotes some parameter associated with a population that we want to estimate. Let $\hat{\theta}$ denote an estimator that will be used to estimate the parameter θ . Then the estimator $\hat{\theta}$ is called an **unbiased** estimator of θ if we have,

$$\mathbb{E} [\hat{\theta}] = \theta$$

, otherwise, if $\mathbb{E}[\hat{\theta}] \neq \theta$, then we say that the estimator is a **biased** estimator.

Definition 2.6 (Bias of an Estimator). For an estimator $\hat{\theta}$, its bias, denoted by $B(\hat{\theta})$, is defined as,

$$B(\hat{\theta}) = \mathbb{E} [\hat{\theta}] - \theta$$

Definition 2.7 (Mean Square Error of an Estimator). For an estimator $\hat{\theta}$, its Mean Square Error, denoted by $MSE(\hat{\theta})$, is defined as,

$$MSE(\hat{\theta}) = \mathbb{E} \left[(\hat{\theta} - \theta)^2 \right]$$

Exercise 2.2. Prove that $MSE(\hat{\theta}) = Var(\hat{\theta}) + (B(\hat{\theta}))^2$

Definition 2.8 (Standard Error). The standard deviation of an estimator $\hat{\theta}$, $SD(\hat{\theta})$, is called standard error of $\hat{\theta}$. Recall that,

$$SD(\hat{\theta}) = \sqrt{Var(\hat{\theta})}$$

3 Some Common Estimators

In this section, we will see common estimators.

3.1 Sample Mean

Suppose we want to estimate mean of some population. To do so, the best that we can do is use some (random) sample and use that to find an estimate for the mean.

Example. Suppose we are interested in determining the average amount of time per day spent by peoples of a large community on the Internet. Now if we want to determine this mean *exactly*, then we will have to collect the information for each individual that belongs to this community, i.e., we will need to have access to the whole population. For all practical purposes we assume that such a task is impossible. So instead we collect some *appropriately* sized sample and based on this we will try to estimate the average time spent by peoples of this community on the Internet.

Suppose we collect information for some n people. Let X_1 denote the amount of time the first person spends on the Internet. Similarly, let X_i denote the amount of time the i_{th} person spends on the Internet. Now observe that these variables $X_1, \dots, X_n$ are in fact, random variables. But let's be careful with our claim. Let's try to see precisely why these variables are random variables. Consider X_1 . It denotes the amount of time spent by the first person on the Internet. Now X_1 is a random variable so it means that it is a function from the associated sample space to the set of all real numbers. What will be the sample space in this case? Observe that when we inquire the first person about the amount of time he/she spends on the Internet, then the reply can be any number between 0-24 hours. This is the sample space, the closed interval $[0, 24]$ and thus the associated mapping from the sample space to the set of real numbers is an identity mapping. Therefore, X_1 is a random variable and similarly you can convince yourself that all other X_i 's are random variables.

Remark. We will assume throughout that these random variables $X_1, X_2, \dots, X_n$ are iid, i.e., they are independent and identically distributed. Although, if you think about it, there is really no possible way that we can prove such a thing. But this assumption makes all the computations and analysis easier than it would be otherwise. Because we want to estimate population mean, if we assume that all these random variables X_i are identically distributed then it means that they all have the same mean, say, μ . So we are trying to estimate this μ .

Now intuitively, one approach to estimate population mean can be to consider the collected sample, and compute its arithmetic mean, i.e., sum all the observations and divide by the total number of observations. But then the question is, how do we know that such an estimate is a *good* estimate for the population mean?

The following theorem explains why the so called sample mean is an unbiased estimator for the population mean.

Theorem 3.1. Let $X_1, X_2, \dots, X_n$ be n iid random variables, where $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$, for $i \in [n]$. Then the random variable $\bar{X}$, called Sample Mean, defined as,

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

is such that, $\mathbb{E}[\bar{X}] = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n$.

Proof. First, let's proof that the $\mathbb{E}[\bar{X}] = \mu$.

$$\begin{aligned}
\mathbb{E}[\bar{X}] &= \mathbb{E}\left[\frac{X_1 + X_2 + \cdots + X_n}{n}\right] \\
&= \frac{1}{n} \mathbb{E}[X_1 + X_2 + \cdots + X_n] && (\mathbb{E}[aX + b] = a\mathbb{E}[X] + b) \\
&= \frac{1}{n} (\mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n]) && (\text{Theorem 1.1}) \\
&= \mu && \mathbb{E}[X_i] = \mu
\end{aligned}$$

Now, we proof that the $\text{Var}(\bar{X}) = \sigma^2/n$.

$$\begin{aligned}
\text{Var}(\bar{X}) &= \frac{1}{n^2} (\text{Var}(X_1 + X_2 + \cdots + X_n)) && (\text{Var}(aX + b) = a^2\text{Var}(X)) \\
&= \frac{1}{n^2} (\text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)) && (\text{Theorem 1.3}) \\
&= \sigma^2/n && (\text{Var}(X_i) = \sigma^2)
\end{aligned}$$

□

The following two corollary follows directly from the above theorem.

Corollary 3.1. Sample Mean, $\bar{X}$, is an unbiased estimator, i.e., $\mathbb{E}[\bar{X}] = \mu$ for the population mean.

Corollary 3.2. Standard Error of a sample mean, $SD(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sigma/\sqrt{n}$.

3.2 Sample Variance

When we wanted to estimate population mean, our intuition suggested considering *sample mean*, i.e., collect some sample and compute its (arithmetic) mean and declare that as an estimate for the population mean. In the preceeding subsection we actually were able to show that the sample mean is an unbiased estimator. Now suppose, we want to estimate the population variance. Then as per our intuition we would be inclined on using the following estimator, say T^2 , defined as,

$$T^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

But we saw in the Practical 1 that the above estimator is not really an unbiased estimator. The following theorem explains why dividing by $n - 1$ yields an unbiased estimator.

Theorem 3.2. Let $X_1, X_2, \cdots, X_n$ be n iid random variables, where $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$, for $i \in [n]$. Then the sample variance, S^2 , defined as,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is an unbiased estimator for σ^2 , i.e., $\mathbb{E}[S^2] = \sigma^2$.

Proof. The following equations will be useful in the proof,

$$\begin{aligned}\mathbb{E}[X_i^2] &= \text{Var}(X_i) + (\mathbb{E}[X_i])^2 && \text{(Definition of Variance)} \\ &= \sigma^2 + \mu^2 && (1)\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\bar{X}^2] &= \text{Var}(\bar{X}) + (\mathbb{E}[\bar{X}])^2 && \text{(Definition of Variance)} \\ &= \sigma^2/n + \mu^2 && (2)\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X_i\bar{X}] &= \mathbb{E}\left[\frac{X_iX_1 + \dots + X_i^2 + \dots + X_iX_n}{n}\right] \\ &= \frac{1}{n}(\mathbb{E}[X_iX_1] + \dots + \mathbb{E}[X_i^2] + \dots + \mathbb{E}[X_iX_n]) && \text{(Why?)} \\ &= \frac{1}{n}((n-1)\mu^2 + \text{Var}(X_i) + (\mathbb{E}[X_i])^2) && \text{(Theorem 1.2)} \\ &= \frac{1}{n}((n-1)\mu^2 + \sigma^2 + \mu^2) && \text{(Equation 1)} \\ &= \sigma^2/n + \mu^2 && (3)\end{aligned}$$

Now, for the actual proof,

$$\begin{aligned}\mathbb{E}[S^2] &= \mathbb{E}\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\ &= \frac{1}{n-1} \mathbb{E}\left[\sum_{i=1}^n (X_i - \bar{X})^2\right] && \text{(Why?)} \\ &= \frac{1}{n-1} \mathbb{E}\left[\sum_{i=1}^n (X_i^2 + \bar{X}^2 - 2X_i\bar{X})\right] \\ &= \frac{1}{n-1} \sum_{i=1}^n \mathbb{E}[X_i^2] + \mathbb{E}[\bar{X}^2] - \mathbb{E}[2X_i\bar{X}] && \text{(Why?)} \\ &= \frac{1}{n-1} \sum_{i=1}^n \sigma^2 + \mu^2 + \sigma^2/n + \mu^2 - 2\mathbb{E}[X_i\bar{X}] && \text{(Equation 1 and 2)} \\ &= \frac{1}{n-1} \sum_{i=1}^n \sigma^2 + \mu^2 + \sigma^2/n + \mu^2 - (2(\sigma^2/n + \mu^2)) && \text{(Equation 3)} \\ &= \frac{1}{n-1} \sum_{i=1}^n \frac{n-1}{n} \sigma^2 \\ &= \sigma^2\end{aligned}$$

□

Therefore, sample variance, S^2 , as defined above in the theorem is an unbiased estimator for the variance.

Exercise 3.1. Consider following quantity, T^2 , defined as,

$$T^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

Show that T^2 is a biased estimator for the population variance and find its bias.

3.3 Estimating Binomial Parameter p

Suppose we have a binomial random variable $Y \sim \text{Bin}(n, p)$. In this section, we will learn how to estimate the parameter p for a binomial random variable. To get a sense of what it is exactly that we are trying to estimate, Assume that we have a coin. We can never determine the true probability of the coin landing on heads but here we are going to estimate this probability based on some collected sample, which in this case will be tossing the given coin some n number of times and counting the number of heads occurred.

Now suppose we toss the given coin some n number of times and we observe y_0 number of heads in total. Then we have the following proposition.

Proposition 3.1. Let $Y \sim \text{Bin}(n, p)$ be a binomial random variable. Then for any $y_0 \in \{0, 1, \dots, n\}$, the following quantity

$$\Pr(Y = y_0) = \binom{n}{y_0} p^{y_0} (1 - p)^{n - y_0}$$

is *maximized* at $p = y_0/n$.

Now based on the above proposition, we have the following theorem,

Theorem 3.3. Let $Y \sim \text{Bin}(n, p)$. Then $\hat{p}$ is an unbiased estimator for the parameter p , i.e., $\mathbb{E}[\hat{p}] = p$, where $\hat{p}$ is defined as,

$$\hat{p} = \frac{Y}{n}$$

Proof. We did this in the class and those who were present, they even wrote the entire proof in their notebooks. $\square$

The following corollary follows from the above theorem and once again its proof was also written by those who were present in the corresponding lecture.

Corollary 3.3. Standard error of estimator $\hat{p}$,

$$SD(\hat{p}) = \sqrt{\frac{pq}{n}},$$

where, $q = 1 - p$.

4 Evaluating Goodness of an Estimator

An estimator can be *good* or *bad*, based on the definition of this goodness and badness. So far we have been calling an estimator good, if it is an unbiased estimator. In this section, we will see how to get even stronger guarantees for the goodness of an estimator.

Definition 4.1 (Error of Estimation). The absolute difference between an estimator and its target parameter is called an Error of Estimation. We denote it by, $\mathcal{E}$, and it is defined as,

$$\mathcal{E} = |\hat{\theta} - \theta|$$

Clearly, this error of estimation, $\mathcal{E}$ is a random quantity, and in fact, we can never compute it exactly. So what we tend to do is try to get some probabilistic bound on this value $\mathcal{E}$, i.e., something of the form,

$$\Pr(\mathcal{E} < b) = r$$

where, b is a real number and $0 \leq r \leq 1$.

The following example would help us understand this better.

Example. Suppose an (unbiased) estimator $\hat{\theta}$ has an approximately normal distribution. Now we know that if we have a normal random variable $X \sim \text{Norm}(\mu, \sigma)$, then,

$$\Pr(|X - \mu| \leq 2\sigma) \approx 0.95$$

Now, since $\hat{\theta} \sim \text{Norm}(\theta, SD(\hat{\theta}))$ (Why?), we have that,

$$\begin{aligned} \Pr(|\hat{\theta} - \theta| \leq 2 \cdot SD(\hat{\theta})) &= 0.95 \\ \Pr(\mathcal{E} \leq 2 \cdot SD(\hat{\theta})) &= 0.95 \end{aligned} \tag{4}$$

The above example suggests that it is reasonable to ask what is the probability of the error of estimation being within a distance of 2 standard deviations for an unbiased estimator $\hat{\theta}$. This is otherwise called, a 2-standard-error bound on the estimator.

The following example taken from the suggested textbook, Mathematical Statistics with Application by Dennis Wackerly and others, will be even more helpful to understand how to use this concept.

Example. Suppose we sample some $n = 1000$ voters and it turns out that 560 of them favor the candidate Jones in an election. Find an estimate of the true proportion of the population who favor Jones and place a 2 standard error bound on your estimate.

Let $Y \sim \text{Bin}(n = 1000, p)$, which denotes the total number of people that favor Jones in a sample of 1000. Then the estimator $\hat{p}$ for p , tells us that 560/1000, or the probability of a randomly selected person favoring Jones is 0.56. Now we know that a binomial random variable $Y \sim \text{Bin}(n = 1000, p)$, Y/n tends to have approximately normal distribution for relatively large values of n (Refer to the last section of the notes Essentials of Probability Theory). In this case, the sample size of 1000 is indeed appropriately large and so the random variable Y will have an approximately normal distribution. So, we have, $\hat{p} \sim \text{Norm}(p, \sqrt{pq/n})$.

Now to compute $SD(\hat{p})$, we would need to know the value of p , but that is exactly what we are trying to estimate. We get around this problem by substituting value of $\hat{p}$ itself in place of p , i.e., we consider, $p = \hat{p}$ and $q = 1 - \hat{p}$. With the above substitution, we get, $2 \cdot SD(\hat{p}) = 0.03$. Then from Equation 4 we have,

$$\begin{aligned} \Pr(\mathcal{E} \leq 2 \cdot SD(\hat{p})) &= 0.95 \\ \Pr(|\hat{p} - p| \leq 0.03) &= 0.95 \end{aligned}$$

The last equation above implies that with probability of 0.95, our estimate of 0.56 has an error of just 0.03. This tells us that our estimation is quite good and Jones is very likely to win the elections.